



Computer Engineering Department
Computer Engineering
IC6400 - Operations Research

Project 0

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$$\begin{aligned}
& \text{Max } Z = 7x_1 + 16x_2 + 16x_3 + 13x_4 + 2x_5 + 8x_6 \\
& \text{such that : } 2x_1 + 3x_2 + 2x_3 + 4x_4 + 2x_5 + 3x_6 \leq 15 \\
& x_1, x_2, x_3, x_4, x_5, x_6 \in \{0, 1\}
\end{aligned}$$

	Items					
Knapsack	x_1	x_2	x_3	x_4	x_5	x_6
0	0	0	0	0	0	0
1	0	0	0	0	0	0
2	7	7	16	16	16	16
3	7	16	16	16	16	16
4	7	16	23	23	23	23
5	7	23	32	32	32	32
6	7	23	32	32	32	32
7	7	23	39	39	39	39
8	7	23	39	39	39	40
9	7	23	39	45	45	45
10	7	23	39	45	45	47
11	7	23	39	52	52	52
12	7	23	39	52	52	53
13	7	23	39	52	54	54
14	7	23	39	52	54	60
15	7	23	39	52	54	60

$$Z = 60, x_1 = 1, x_2 = 1, x_3 = 1, x_4 = 1, x_5 = 0, x_6 = 1$$

	Items					
Knapsack	x_2	x_3	x_4	x_6	x_1	x_5
Cost	3	2	4	3	2	2
Value	16	16	13	8	7	2

$$Z = 60, x_2 = 1, x_3 = 1, x_4 = 1, x_6 = 1, x_1 = 1, x_5 = 0$$

	Items					
Knapsack	x_3	x_2	x_4	x_1	x_6	x_5
Cost	2	3	4	2	3	2
Value	16	16	13	7	8	2
Proportion	8.000000	5.333333	3.250000	3.500000	2.666667	1.000000

$$Z = 60, x_3 = 1, x_2 = 1, x_4 = 1, x_1 = 1, x_6 = 1, x_5 = 0$$